

Week 05 — Trends, Seasonality & SARIMA

19 March 2026

The big picture. Make a series stationary by removing trend and seasonality through differencing; ARIMA and SARMA encode this in the polynomials.

Real series are almost never stationary: they *drift* (a trend) and they *repeat* (a season). The working model of the chapter is $X_t = \mu_t + Y_t$, with μ_t a deterministic time-dependent mean and $\{Y_t\}$ stationary. There are two ways to peel μ_t off: **estimate-and-subtract** (regress on t and on seasonal dummies) or **difference it away** with $\nabla = I - B$. Layering AR/MA structure onto both ordinary and seasonal lags gives the SARMA family; folding the differencing in front gives (S)ARIMA — the Box–Jenkins workhorse.

Key definitions

Definition — Mean-trend (signal-plus-noise) model

$X_t = \mu_t + Y_t$, where μ_t is a deterministic, time-dependent mean (the **trend**) and $\{Y_t\}$ is zero-mean second-order stationary. Canonical cases: $\mu_t = a + bt$ (linear), μ_t a degree- $(q-1)$ polynomial, or $\mu_t = s_t$ **periodic** of period s ($s_{t+s} = s_t$). The two can coexist: $X_t = a + bt + s_t + Y_t$.

Definition — Difference operators ∇ and $\nabla_{(s)}$

Ordinary difference: $\nabla = I - B$, $\nabla X_t = X_t - X_{t-1}$ (with $BX_t = X_{t-1}$). It is linear, and

$$\nabla^d = (I - B)^d = \sum_{k=0}^d \binom{d}{k} (-1)^k B^k, \quad \text{e.g. } \nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}.$$

Seasonal difference of period s : $\nabla_{(s)} = I - B^s$, $\nabla_{(s)} X_t = X_t - X_{t-s}$ (just two terms). Ordinary is the case $s = 1$.

Definition — Seasonal MA and AR (SMA, SAR)

Only lags that are *multiples of s* appear:

$$\text{SMA}(Q)_s: \quad X_t = \varepsilon_t - \sum_{j=1}^Q \theta_j^{(s)} \varepsilon_{t-js} = \Theta^{(s)}(B) \varepsilon_t,$$

$$\text{SAR}(P)_s: \quad \Phi^{(s)}(B) X_t = \varepsilon_t, \quad X_t = \varepsilon_t + \sum_{j=1}^P \phi_j^{(s)} X_{t-js}.$$

Polynomials: $\Theta^{(s)}(z) = 1 - \sum_j \theta_j^{(s)} z^{sj}$ and $\Phi^{(s)}(z) = 1 - \sum_j \phi_j^{(s)} z^{sj}$. Basic seasonal MA(1): $X_t = \varepsilon_t - \theta \varepsilon_{t-s}$.

Definition — SARMA(p, q) $\times$ (P, Q)_s

Multiply an ordinary ARMA by a seasonal ARMA at period s :

$$\Phi(B) \Phi^{(s)}(B) X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t.$$

$\text{SMA}(Q)_s = \text{SARMA}(0, 0) \times (0, Q)_s$ and $\text{SAR}(P)_s = \text{SARMA}(0, 0) \times (P, 0)_s$. Example, $\text{SARMA}(0, 1) \times (0, 1)_{12}$: $X_t = (1 - \theta B)(1 - \theta^{(12)} B^{12}) \varepsilon_t$.

Definition — ARIMA(p, d, q) and SARIMA(p, d, q) $\times$ (P, D, Q) $_s$

$\{X_t\}$ is **ARIMA**(p, d, q) (“integrated ARMA”) if its d -th difference is a causal, invertible ARMA(p, q):

$$\Phi(B) (1 - B)^d X_t = \Theta(B) \varepsilon_t, \quad \text{i.e. } Z_t = \nabla^d X_t \text{ is ARMA}(p, q).$$

The “I” is *integrated*: recover X_t from Z_t by summing d times. The full **SARIMA** folds in both an ordinary difference d and a seasonal difference D :

$$\Phi(B) \Phi^{(s)}(B) (1 - B)^d (1 - B^s)^D X_t = \Theta(B) \Theta^{(s)}(B) \varepsilon_t.$$

Definition — Random walk, IMA(1, 1), ARI(1, 1)

Random walk $X_t = X_{t-1} + \varepsilon_t$ is the AR(1) at $\phi = 1$: *not* stationary, but $\nabla X_t = \varepsilon_t$ is white noise, so it is ARIMA(0, 1, 0). Two minimal one-step-integrated models, both with non-stationary $\{Y_t\}$:

- **IMA**(1, 1)=ARIMA(0, 1, 1): $Y_t = Y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$ (i.e. $\nabla Y_t = \varepsilon_t - \theta \varepsilon_{t-1}$);
- **ARI**(1, 1)=ARIMA(1, 1, 0): $Y_t - Y_{t-1} = \phi(Y_{t-1} - Y_{t-2}) + \varepsilon_t$, $|\phi| < 1$.

Subtract vs. difference

Regress-and-subtract ($a + bt$, or sin / cos harmonics) gives an interpretable deterministic trend, but you must *know its form*. **Differencing** is model-free and robust: ∇ kills any linear drift, ∇^q any degree- $(q - 1)$ polynomial, $\nabla_{(s)}$ any period- s season — at the price of slightly distorting the noise (∇Y_t replaces Y_t) and risking *over*-differencing.

Key results & formulas

Differencing removes a polynomial trend

If $X_t = a + bt + Y_t$ with $\{Y_t\}$ stationary, then $\nabla X_t = b + \nabla Y_t$: the trend collapses to the constant b , and ∇Y_t is again stationary. More generally, if μ_t is a degree- $(q - 1)$ polynomial then $\nabla^q \mu_t \equiv 0$, so

$$\nabla^q X_t = \nabla^q Y_t = \sum_{k=0}^q \binom{q}{k} (-1)^k Y_{t-k}.$$

Idea: each ∇ lowers a polynomial’s degree by one; linearity passes through to Y .

How many differences?

∇ lowers a polynomial degree by exactly one. So **linear drift** $\Rightarrow d = 1$, **quadratic** $\Rightarrow d = 2$: choose $d = q$ to flatten a degree- $(q - 1)$ trend. A seasonal pattern of period s needs *one* seasonal difference $\nabla_{(s)}$ ($D = 1$).

Theorem — $\nabla \nabla_{(12)}$ flattens trend + season

Let $X_t = a + bt + s_t + Y_t$ with $s_{t+12} = s_t$ and $\{Y_t\}$ zero-mean stationary. Apply the seasonal difference first ($\nabla_{(12)} X_t = 12b + \nabla_{(12)} Y_t$ kills s_t and turns bt into a constant), then ∇ kills the constant:

$$\nabla \nabla_{(12)} X_t = \nabla \nabla_{(12)} Y_t = Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}.$$

This is a fixed linear combination of stationary Y ’s: mean 0, covariance a function of τ only $\Rightarrow$ weakly stationary. (This is the standard “ ∇_{12} then ∇ ” pre-processing for monthly data, e.g. the Mauna Loa CO₂ series.)

Multiplicative season needs a different recipe

If trend and season *multiply*, $X_t = (a + bt)s_t + Y_t$ (amplitude grows with level), then $\nabla \nabla_{(12)}$ is *wrong*: here a single *double seasonal* difference works, $\nabla_{(12)}^2 X_t = Y_t - 2Y_{t-12} + Y_{t-24}$, again stationary.

ACVS of the seasonal MA(1)

For $X_t = \varepsilon_t - \theta \varepsilon_{t-s}$ (white-noise variance σ_ε^2), only three lags survive:

$$\gamma_0 = (1 + \theta^2)\sigma_\varepsilon^2, \quad \gamma_{\pm s} = -\theta \sigma_\varepsilon^2, \quad \gamma_\tau = 0 \text{ else} \quad \Rightarrow \quad \rho_{\pm s} = \frac{-\theta}{1 + \theta^2}.$$

Idea: white-noise filter rule $\gamma_\tau = \sigma_\varepsilon^2 \sum_j a_j a_{j+|\tau|}$ on taps $a_0 = 1, a_s = -\theta$. A single ACF spike at lag s is the fingerprint of an SMA(1) $_s$.

Random walk and integrated growth (the difference-me signal)

For the random walk $X_t = \sum_{j=1}^t \varepsilon_j$ ($X_0 = 0$): $\mathbb{E}[X_t] = 0$, $\text{Var}(X_t) = t\sigma_\varepsilon^2$, $\text{Cov}(X_t, X_{t+\tau}) = t\sigma_\varepsilon^2$ — all depend on t , so $\{X_t\}$ is *not* stationary, yet $\nabla X_t = \varepsilon_t$ is white noise. Same diagnostic for IMA(1, 1) ($\theta \neq 1$):

$$\text{Var}(Y_t) = \sigma_\varepsilon^2 [1 + (1 - \theta)^2(t - 1) + \theta^2], \quad \text{Corr}(Y_t, Y_{t+\tau}) \rightarrow 1.$$

ARI(1, 1) behaves the same way. **Variance growing linearly in t and nearby values almost perfectly correlated** is the signature that a difference is needed.

Theorem (pitfall) — $\nabla_{(s)} \neq \nabla^s$, and over-differencing

$\nabla_{(s)} = I - B^s$ subtracts the value s steps back (two terms, removes a period- s season); $\nabla^s = (I - B)^s$ is an $(s + 1)$ -term binomial sum (removes a degree- $(s - 1)$ polynomial). They are not the same. Differencing an already-stationary series injects a $(1 - B)$ factor into the MA part: a non-invertible MA(1) with $\rho_1 \rightarrow -\frac{1}{2}$. Seeing $\rho_1 \approx -\frac{1}{2}$ after differencing means *difference one time fewer*.

Seasonal ARMA $(1 - 0.7B^2)X_t = (1 - 0.3B^2)\varepsilon_t$

Write $\phi = 0.7, \theta = 0.3$. AR roots have modulus $> 1 \Rightarrow$ causal; expand the geometric series in B^2 . Only *even* lags survive: $a_0 = 1, a_j = 0$ for odd j , and $a_j = \phi^{(j-2)/2}(\phi - \theta)$ for even $j \geq 2$. Hence odd-lag correlations vanish and the ACF decays geometrically at rate ϕ along even lags: $\rho_2 \approx 0.47, \rho_4 \approx 0.33, \rho_6 \approx 0.23$ — the correlogram shows spikes only at $\tau = 2, 4, 6, \dots$ decaying like $0.7^{\tau/2}$.

Intuition

The whole chapter in one line

Strip non-stationarity *before* modelling: ∇^d for trend, $\nabla_{(s)}^D$ for season; then fit a SARMA on the (now stationary) result. SARIMA bundles all of this into four polynomials plus two differencing exponents.

Reading a correlogram

Ordinary ARMA structure lives at small lags $1, 2, 3, \dots$; **seasonal** structure clusters around $s, 2s, \dots$ (with small side-lobes at $s \pm 1$ from cross terms). A slowly decaying sample ACF $\Rightarrow$ a unit root $\Rightarrow$ difference. A lone $-\frac{1}{2}$ at lag 1 $\Rightarrow$ you over-differenced.

Key takeaways

Key takeaways — the week's reflexes

- **Recipe.** $X_t = \mu_t + Y_t$: remove μ_t by regression (need the form) *or* differencing (model-free). For monthly data with linear drift: $\nabla\nabla_{(12)}$ (seasonal first, then ordinary).
- **Pick d, D .** $d = q$ flattens a degree- $(q - 1)$ polynomial trend; $D = 1$ removes a period- s season (rarely $D = 2$). Stop when the mean is stable, variance constant, and the ACF decays fast.
- **Know the polynomials.** ARIMA: $\Phi(B)(1 - B)^d X_t = \Theta(B)\varepsilon_t$. SARIMA multiplies in seasonal factors $\Phi^{(s)}, \Theta^{(s)}$ and $(1 - B^s)^D$. Expand by treating B as a formal variable; seasonal factors contribute only multiples of s , cross terms give lags $s \pm 1$.
- **Diagnostics of integration.** Variance growing $\propto t$ and near-unit correlation of nearby points $\Rightarrow$ difference. Random walk, IMA(1, 1), ARI(1, 1) all show this.
- **Traps.** $\nabla_{(s)} \neq \nabla^s$ (season vs. polynomial). Over-differencing creates a non-invertible MA unit root with $\rho_1 \rightarrow -\frac{1}{2}$.